

Australian statement of hazardous nature : Classified as hazardous according to criteria of Safe Work Australia

## Section 1 - Identification

**Product Name** Potassium hydroxide 1M alcoholic

|                                |                                                                                                      |
|--------------------------------|------------------------------------------------------------------------------------------------------|
| <b>Product Code</b>            | J/6610/15, J/6610/17, J/6610/24                                                                      |
| <b>Address</b>                 | ThermoFisher Scientific Australia Pty Ltd<br>5 Caribbean Drive, Scoresby<br>VICTORIA 3179, Australia |
| <b>Emergency Tel.</b>          | <b>CHEMTREC®</b><br><b>03 9757 4559 or +613 9757 4559</b>                                            |
| <b>Telephone / Fax Numbers</b> | Tel: 1300 735 292<br>Fax: 1800 067 639                                                               |
| <b>E-mail address</b>          | ANZinfo@thermofisher.com                                                                             |

**Recommended Use** Laboratory chemicals.

**Uses advised against** This product does not contain any substance(s) on the Illicit Drug Precursors/Reagents list. This product does not contain any substance(s) subject to Prohibition, Authorization or Restriction. This product does not contain any substance(s) listed on the voluntary National Code of Practice for Chemicals of Security Concern.

## Section 2 - Hazard(s) Identification

### Classification under Safe Work Australia

Classified as hazardous according to criteria of Safe Work Australia

#### Physical hazards

Flammable liquids

Category 2

#### Health hazards

Skin Corrosion/Irritation  
Serious Eye Damage/Eye Irritation  
Specific target organ toxicity - (single exposure)

Category 1 A  
Category 1  
Category 1

#### Environmental hazards

No hazards identified

### Label Elements



Flame



Health Hazard



Corrosion

**Signal Word****Danger****Hazard Statements**

H225 - Highly flammable liquid and vapor  
H314 - Causes severe skin burns and eye damage  
H370 - Causes damage to organs  
AUH071 - Corrosive to the respiratory tract

**Precautionary Statements**

P201 - Obtain special instructions before use  
P202 - Do not handle until all safety precautions have been read and understood  
P210 - Keep away from heat, hot surfaces, sparks, open flames and other ignition sources. No smoking  
P233 - Keep container tightly closed  
P240 - Ground and bond container and receiving equipment  
P242 - Use non-sparking tools  
P243 - Take action to prevent static discharges  
P260 - Do not breathe dust/fume/gas/mist/vapors/spray  
P264 - Wash face, hands and any exposed skin thoroughly after handling  
P270 - Do not eat, drink or smoke when using this product  
P271 - Use only outdoors or in a well-ventilated area  
P280 - Wear protective gloves/protective clothing/eye protection/face protection  
P301 + P330 + P331 - IF SWALLOWED: rinse mouth. Do NOT induce vomiting  
P303 + P361 + P353 - IF ON SKIN (or hair): Take off immediately all contaminated clothing. Rinse skin with water or shower  
P304 + P340 - IF INHALED: Remove person to fresh air and keep comfortable for breathing  
P305 + P351 + P338 - IF IN EYES: Rinse cautiously with water for several minutes. Remove contact lenses, if present and easy to do. Continue rinsing  
P310 - Immediately call a POISON CENTER or doctor  
P363 - Wash contaminated clothing before reuse  
P370 + P378 - In case of fire: Use dry sand, dry chemical or alcohol-resistant foam to extinguish  
P403 + P233 - Store in a well-ventilated place. Keep container tightly closed  
P501 - Dispose of contents/ container to an approved waste disposal plant

**Other information****Section 3 - Composition and Information on Ingredients**

| Component           | CAS No    | Weight % |
|---------------------|-----------|----------|
| Ethyl alcohol       | 64-17-5   | 84.4     |
| Potassium hydroxide | 1310-58-3 | 5.6      |

**Section 4 - First Aid Measures**

|                                           |                                                                                                                                                                                                                                                                                                                         |
|-------------------------------------------|-------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|
| <b>Inhalation</b>                         | Remove to fresh air. If breathing is difficult, give oxygen. Do not use mouth-to-mouth method if victim ingested or inhaled the substance; give artificial respiration with the aid of a pocket mask equipped with a one-way valve or other proper respiratory medical device. Immediate medical attention is required. |
| <b>Ingestion</b>                          | Do NOT induce vomiting. Call a physician or poison control center immediately.                                                                                                                                                                                                                                          |
| <b>Skin Contact</b>                       | Wash off immediately with plenty of water for at least 15 minutes. Immediate medical attention is required.                                                                                                                                                                                                             |
| <b>Eye Contact</b>                        | Rinse immediately with plenty of water, also under the eyelids, for at least 15 minutes. Immediate medical attention is required. Do not rub affected area.                                                                                                                                                             |
| <b>Self-Protection of the First Aider</b> | Ensure that medical personnel are aware of the material(s) involved, take precautions to                                                                                                                                                                                                                                |

protect themselves and prevent spread of contamination.

**First Aid Facilities**

Eyewash, safety shower and washroom.

**Most important symptoms and effects**

Difficulty in breathing. Causes burns by all exposure routes. Ingestion causes severe swelling, severe damage to the delicate tissue and danger of perforation: Inhalation of high vapor concentrations may cause symptoms like headache, dizziness, tiredness, nausea and vomiting

**Notes to Physician**

Treat symptomatically. Symptoms may be delayed.

## Section 5 - Fire Fighting Measures

**Suitable Extinguishing Media**

CO<sub>2</sub>, dry chemical, dry sand, alcohol-resistant foam. Water spray. Water mist may be used to cool closed containers.

**Extinguishing media which must not be used for safety reasons**

Do not use a solid water stream as it may scatter and spread fire.

**Hazardous Decomposition Products**

Carbon monoxide (CO), Carbon dioxide (CO<sub>2</sub>), Thermal decomposition can lead to release of irritating gases and vapors.

**Specific Hazards Arising from the Chemical**

Flammable. Containers may explode when heated. Vapors may form explosive mixtures with air. Vapors may travel to source of ignition and flash back.

**Special protective equipment and precautions for fire fighters**

As in any fire, wear self-contained breathing apparatus pressure-demand, MSHA/NIOSH (approved or equivalent) and full protective gear. Thermal decomposition can lead to release of irritating gases and vapors.

## Section 6 - Accidental Release Measures

**Emergency procedures**

Use personal protective equipment as required. Keep people away from and upwind of spill/leak. Evacuate personnel to safe areas. Remove all sources of ignition. Take precautionary measures against static discharges.

**Environmental Precautions**

Should not be released into the environment. Do not flush into surface water or sanitary sewer system. See Section 12 for additional Ecological Information.

**Methods for Containment and Clean Up****Clean-up methods - small spillage**

Soak up with inert absorbent material. Keep in suitable, closed containers for disposal. Remove all sources of ignition. Use spark-proof tools and explosion-proof equipment.

**Clean-up methods - large spillage**

Typically only supplied in small quantities as packaged goods.

If extremely toxic or used in larger quantities ensure a spillage action plan is in place. Evacuate area. Control the source and/or contain the spill if safe and able to do so. Use temporary diking, sand bags, dry sand, earth or proprietary booms/absorbent pads if available. Obtain advice on containment, neutralisation and clean-up from local emergency responders.

**Reference to Other Sections**

Refer to protective measures listed in Sections 8 and 13.

## Section 7 - Handling and Storage

**Precautions for Safe Handling**

Use only under a chemical fume hood. Wear personal protective equipment/face protection. Do not get in eyes, on skin, or on clothing. Keep away from open flames, hot surfaces and sources of ignition. Use only non-sparking tools. Use spark-proof tools and explosion-proof equipment. Do not breathe (dust, vapor, mist, gas). Do not ingest. If swallowed then seek immediate medical assistance. Take precautionary measures against static discharges. Wash hands before breaks and immediately after handling the product. To avoid ignition of vapors by static electricity discharge, all metal parts of the equipment must be grounded.

**Conditions for Safe Storage, Including any Incompatibilities**

Keep containers tightly closed in a dry, cool and well-ventilated place. Corrosives area. Flammables area. Keep away from heat, sparks and flame.

AS/NZS 2243.10:2004, Safety in laboratories - Storage of chemicals

AS 1940-2004 - The storage and handling of flammable and combustible liquids

## Section 8 - Exposure Controls and Personal Protection

**Exposure limits**

**AUS** - Exposure Standards for Atmospheric Contaminants in the Occupational Environment - Guidance Note on the Interpretation of Exposure Standards for Atmospheric Contaminants in the Occupational Environment [NOHSC:3008(1995)]

Adopted National Exposure Standards for Atmospheric Contaminants in the Occupational Environment [NOHSC:1003(1995)]

updated in August, 2005. Safe Work Australia **ACGIH** - Threshold Limit Values - Ceiling (TLV-C) guidelines by the American Conference of Governmental Industrial Hygienists (ACGIH) for controlling worker exposure to airborne chemical concentrations in the workplace. **UK** - EH40/2005 Work Exposure Limits, Fourth edition. Published 2020. **DE** - MAK and BAT values of Hazardous Chemical Compounds in the Work Area. Published by German Research Foundation on July 1, 2011

| Component           | Australia                                    | New Zealand WEL                              | ACGIH TLV                    | The United Kingdom                                                                                               | Germany                                           |
|---------------------|----------------------------------------------|----------------------------------------------|------------------------------|------------------------------------------------------------------------------------------------------------------|---------------------------------------------------|
| Ethyl alcohol       | TWA: 1000 ppm<br>TWA: 1880 mg/m <sup>3</sup> | TWA: 1000 ppm<br>TWA: 1880 mg/m <sup>3</sup> | STEL: 1000 ppm               | TWA: 1000 ppm TWA;<br>1920 mg/m <sup>3</sup> TWA<br>WEL - STEL: 3000 ppm<br>STEL; 5760 mg/m <sup>3</sup><br>STEL | 200 ppm TWA MAK;<br>380 mg/m <sup>3</sup> TWA MAK |
| Potassium hydroxide | Ceiling: 2 mg/m <sup>3</sup>                 | Ceiling: 2 mg/m <sup>3</sup>                 | Ceiling: 2 mg/m <sup>3</sup> | WEL - 2 mg/m <sup>3</sup> STEL                                                                                   |                                                   |

**Biological limit values**

**NZ** - Substances assigned Biological Exposure Indices in the New Zealand Workplace Exposure Standards and Biological Exposure Indices (6th edition). New Zealand Department of Labor

**Exposure Controls****Engineering Measures**

Use only under a chemical fume hood. Ensure that eyewash stations and safety showers are close to the workstation location. Use explosion-proof electrical/ventilating/lighting equipment. Ensure adequate ventilation, especially in confined areas.

Wherever possible, engineering control measures such as the isolation or enclosure of the process, the introduction of process or equipment changes to minimise release or contact, and the use of properly designed ventilation systems, should be adopted to control hazardous materials at source

**Personal protective equipment****Eye Protection**

Goggles (Australian/New Zealand Standard AS/NZS 1337 - Eye protectors for Industrial applications)

**Hand Protection**

Protective gloves

| Glove material | Breakthrough time | Glove thickness | AUS/NZ Standard | Glove comments        |
|----------------|-------------------|-----------------|-----------------|-----------------------|
| Viton (R)      | > 480 minutes     | 0.7mm           | AS/NZS 2161     | (minimum requirement) |

Inspect gloves before use.

Please observe the instructions regarding permeability and breakthrough time which are provided by the supplier of the gloves. (Refer to manufacturer/supplier for information)

Ensure gloves are suitable for the task: Chemical compatibility, Dexterity, Operational conditions, User susceptibility, e.g. sensitisation effects, also take into consideration the specific local conditions under which the product is used, such as the danger of cuts, abrasion.

Remove gloves with care avoiding skin contamination.

|                                        |                                                                                                                                                                                                                                                                                                                             |
|----------------------------------------|-----------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|
| <b>Skin and body protection</b>        | Wear appropriate protective gloves and clothing to prevent skin exposure; Impervious clothing                                                                                                                                                                                                                               |
| <b>Respiratory Protection</b>          | Use an AS/NZS 1716 approved respirator if exposure limits are exceeded or if irritation or other symptoms are experienced. To protect the wearer, respiratory protective equipment must be the correct fit and be used and maintained in line with AS/NZS 1715 on the use and maintenance of respiratory protective devices |
| <b>Recommended Filter type:</b>        | low boiling organic solvent Type AX Brown conforming to EN371 or Organic gases and vapours filter Type A Brown conforming to EN14387 (or AUS/NZ equivalent)                                                                                                                                                                 |
| <b>Recommended half mask:-</b>         | Valve filtering: EN405 or Half mask: EN140 plus filter, EN 141 Type AX (or AUS/NZ equivalent)<br>When RPE is used a face piece Fit Test should be conducted                                                                                                                                                                 |
| <b>Hygiene Measures</b>                | Handle in accordance with good industrial hygiene and safety practice.                                                                                                                                                                                                                                                      |
| <b>Environmental exposure controls</b> | Prevent product from entering drains. Do not allow material to contaminate ground water system.                                                                                                                                                                                                                             |

## Section 9 - Physical and Chemical Properties

### Information on basic physical and chemical properties

|                                                |                                               |                                             |
|------------------------------------------------|-----------------------------------------------|---------------------------------------------|
| <b>Appearance</b>                              | Colorless - Light yellow                      |                                             |
| <b>Physical State</b>                          | Liquid                                        |                                             |
| <b>Odor</b>                                    | No information available                      |                                             |
| <b>Odor Threshold</b>                          | No data available                             |                                             |
| <b>pH</b>                                      | > 12                                          |                                             |
| <b>Melting Point/Range</b>                     | No data available                             |                                             |
| <b>Softening Point</b>                         | No data available                             |                                             |
| <b>Boiling Point/Range</b>                     | 78 - 80 °C / 172.4 - 176 °F                   |                                             |
| <b>Flash Point</b>                             | 15 °C / 59 °F                                 | <b>Method -</b> No information available    |
| <b>Evaporation Rate</b>                        | No data available                             |                                             |
| <b>Flammability (solid,gas)</b>                | Not applicable                                | Liquid                                      |
| <b>Explosion Limits</b>                        | <b>Lower</b> 3.3 vol%<br><b>Upper</b> 19 vol% |                                             |
| <b>Vapor Pressure</b>                          | No data available                             |                                             |
| <b>Vapor Density</b>                           | No data available                             | (Air = 1.0)                                 |
| <b>Specific Gravity / Density</b>              | 0.8                                           |                                             |
| <b>Bulk Density</b>                            | Not applicable                                | Liquid                                      |
| <b>Water Solubility</b>                        | Miscible                                      |                                             |
| <b>Solubility in other solvents</b>            | No information available                      |                                             |
| <b>Partition Coefficient (n-octanol/water)</b> |                                               |                                             |
| <b>Component</b>                               | <b>log Pow</b>                                |                                             |
| Ethyl alcohol                                  | -0.32                                         |                                             |
| Potassium hydroxide                            | 0.83                                          |                                             |
| <b>Autoignition Temperature</b>                | 363 °C / 685.4 °F                             |                                             |
| <b>Decomposition Temperature</b>               | No data available                             |                                             |
| <b>Viscosity</b>                               | No data available                             |                                             |
| <b>Explosive Properties</b>                    |                                               | Vapors may form explosive mixtures with air |
| <b>Oxidizing Properties</b>                    | No information available                      |                                             |

### Other information

## Section 10 - Stability and Reactivity

|                   |                                            |
|-------------------|--------------------------------------------|
| <b>Reactivity</b> | None known, based on information available |
| <b>Stability</b>  | Stable under normal conditions.            |

|                                         |                                                                                                                                    |
|-----------------------------------------|------------------------------------------------------------------------------------------------------------------------------------|
| <b>Conditions to Avoid</b>              | Incompatible products, Excess heat, Keep away from open flames, hot surfaces and sources of ignition.                              |
| <b>Incompatible Materials</b>           | Strong oxidizing agents, Acids.                                                                                                    |
| <b>Hazardous Decomposition Products</b> | Carbon monoxide (CO). Carbon dioxide (CO <sub>2</sub> ). Thermal decomposition can lead to release of irritating gases and vapors. |
| <b>Hazardous Polymerization</b>         | Hazardous polymerization does not occur.                                                                                           |

## Section 11 - Toxicological Information

### Information on Toxicological Effects

#### Product Information

|                            |            |
|----------------------------|------------|
| <b>(a) acute toxicity;</b> |            |
| <b>Oral</b>                | Category 4 |
| <b>Dermal</b>              | Category 4 |
| <b>Inhalation</b>          | Category 4 |

#### Toxicology data for the components

| Component           | LD50 Oral                                                    | LD50 Dermal | LC50 Inhalation                                                   |
|---------------------|--------------------------------------------------------------|-------------|-------------------------------------------------------------------|
| Ethyl alcohol       | LD50 = 10470 mg/kg<br>OECD 401 (Rat)<br>3450 mg/kg ( Mouse ) |             | LC50 = 117-125 mg/l (4h)<br>OECD 403 (rat)<br>20000 ppm/10H (rat) |
| Potassium hydroxide | LD50 = 333-384 mg/kg (Rat)                                   |             |                                                                   |

**(b) skin corrosion/irritation;** Category 1 A

**(c) serious eye damage/irritation;** Category 1

**(d) respiratory or skin sensitization;**

|                    |                   |
|--------------------|-------------------|
| <b>Respiratory</b> | No data available |
| <b>Skin</b>        | No data available |

| Component                         | Test method                                    | Test species | Study result    |
|-----------------------------------|------------------------------------------------|--------------|-----------------|
| Ethyl alcohol<br>64-17-5 ( 84.4 ) | Mouse Ear Swelling Test (MEST)                 | mouse        | non-sensitising |
|                                   | OECD Test Guideline 429 Local Lymph Node Assay | mouse        | non-sensitising |

**(e) germ cell mutagenicity;** No data available

| Component                         | Test method                                   | Test species          | Study result |
|-----------------------------------|-----------------------------------------------|-----------------------|--------------|
| Ethyl alcohol<br>64-17-5 ( 84.4 ) | AMES test<br>OECD Test Guideline 471          | in vitro<br>Bacteria  | negative     |
|                                   | Gene cell mutation<br>OECD Test Guideline 476 | in vitro<br>Mammalian | negative     |

**(f) carcinogenicity;** No data available

The table below indicates whether each agency has listed any ingredient as a carcinogen

**(g) reproductive toxicity;** No data available

| Component                         | Test method             | Test species / Duration   | Study result          |
|-----------------------------------|-------------------------|---------------------------|-----------------------|
| Ethyl alcohol<br>64-17-5 ( 84.4 ) | OECD Test Guideline 416 | Oral / mouse 2 Generation | NOAEL = 13.8 g/kg/day |
|                                   | OECD Test Guideline 414 | Inhalation / Rat          | NOAEC = 16000 ppm     |

(h) STOT-single exposure; Category 2

Results / Target organs Optic nerve  
Central nervous system (CNS)

(i) STOT-repeated exposure; No data available

Target Organs No information available.

(j) aspiration hazard; No data available

Symptoms / effects, both acute and delayed Ingestion causes severe swelling, severe damage to the delicate tissue and danger of perforation: Inhalation of high vapor concentrations may cause symptoms like headache, dizziness, tiredness, nausea and vomiting

## Section 12 - Ecological Information

Ecotoxicity effects Contains no substances known to be hazardous to the environment or that are not degradable in waste water treatment plants.

| Component     | Freshwater Fish                                                  | Water Flea                                    | Freshwater Algae                              | Microtox                                                                                                                |
|---------------|------------------------------------------------------------------|-----------------------------------------------|-----------------------------------------------|-------------------------------------------------------------------------------------------------------------------------|
| Ethyl alcohol | Fathead minnow<br>(Pimephales promelas)<br>LC50 = 14200 mg/l/96h | EC50 = 9268 mg/L/48h<br>EC50 = 10800 mg/L/24h | EC50 (72h) = 275 mg/l<br>(Chlorella vulgaris) | Photobacterium<br>phosphoreum: EC50 =<br>34634 mg/L/30 min<br>Photobacterium<br>phosphoreum: EC50 =<br>35470 mg/L/5 min |

### Persistence and Degradability

Persistence Soluble in water, Persistence is unlikely, based on information available.

| Component                         | Degradability   |
|-----------------------------------|-----------------|
| Ethyl alcohol<br>64-17-5 ( 84.4 ) | OECD 301E = 94% |

Bioaccumulative Potential Bioaccumulation is unlikely

| Component           | log Pow | Bioconcentration factor (BCF) |
|---------------------|---------|-------------------------------|
| Ethyl alcohol       | -0.32   | No data available             |
| Potassium hydroxide | 0.83    | No data available             |

Mobility The product is water soluble, and may spread in water systems. : Will likely be mobile in the environment due to its water solubility Highly mobile in soils

Endocrine Disruptor Information This product does not contain any known or suspected endocrine disruptors

Persistent Organic Pollutant This product does not contain any known or suspected substance

Ozone Depletion Potential This product does not contain any known or suspected substance

## Section 13 - Disposal Considerations

Waste from Residues/Unused Products Do not allow into drains or watercourses or dispose of where ground or surface waters may be affected. Wastes, including emptied containers, are controlled wastes and should be disposed of in accordance with all federal, E.P.A., state and local regulations. Assure conformity with all applicable regulations.

Contaminated Packaging Dispose of this container to hazardous or special waste collection point. Empty containers retain product residue, (liquid and/or vapor), and can be dangerous. Keep product and empty container away from heat and sources of ignition.

Other Information Chemical wastes should be disposed through a licensed commercial waste collection service. Do not flush to sewer. Waste codes should be assigned by the user based on the application for which the product was used. Can be landfilled or incinerated, when in compliance with local regulations. Do not empty into drains. Large amounts will affect pH and harm aquatic organisms.

## Section 14 - Transport Information

### IMDG/IMO

**UN-No** UN2924  
**Proper Shipping Name** Flammable liquid, corrosive, n.o.s.  
**Technical Shipping Name** Contains ethanol and potassium hydroxide  
**Hazard Class** 3  
**Subsidiary Hazard Class** 8  
**Packing Group** II

### ADG

**UN-No** UN2924  
**Proper Shipping Name** Flammable liquid, corrosive, n.o.s.  
**Technical Shipping Name** Contains ethanol and potassium hydroxide  
**Hazard Class** 3  
**Subsidiary Hazard Class** 8  
**Packing Group** II

| Component           | Hazchem Code |
|---------------------|--------------|
| Ethyl alcohol       | 2YE          |
| 64-17-5 ( 84.4 )    | 2Y           |
| Potassium hydroxide | 2W           |
| 1310-58-3 ( 5.6 )   | 2R           |

### IATA

**UN-No** UN2924  
**Proper Shipping Name** Flammable liquid, corrosive, n.o.s.  
**Technical Shipping Name** Contains ethanol and potassium hydroxide  
**Hazard Class** 3  
**Subsidiary Hazard Class** 8  
**Packing Group** II

**Environmental hazards** No hazards identified

**Special Precautions** No special precautions required

**Additional information** None known

## Section 15 - Regulatory Information

### Safety, health and environmental regulations/legislation specific for the substance or mixture

#### National Regulations Australia

See section 8 for national exposure control parameters.

#### Standard for the Uniform Scheduling of Medicines and Poisons

Classified as a scheduled poison according to the Standard for Uniform Scheduling of Medicines and Poisons.

| Component                       | Standard for the Uniform Scheduling of Medicines and Poisons                                                                                                                                                                                                                                                                                                                                                                                                                                           |
|---------------------------------|--------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|
| Potassium hydroxide - 1310-58-3 | <p>Schedule 5 listed - except its salts and derivatives; in preparations being: solid preparations the pH of which in a 10 g/L aqueous solution is &gt;11.5; liquid or semi-solid preparations the pH of which is &gt;11.5 except in food additive preparations for domestic use</p> <p>Schedule 6 listed - except its salts and derivatives; except: [a] when included in Schedule 5 or Schedule 10, [b] in preparations containing &lt;=5% of Potassium hydroxide being: [i] solid preparations,</p> |

|  |                                                                                                                                                                |
|--|----------------------------------------------------------------------------------------------------------------------------------------------------------------|
|  | the pH of which in a 10 g/L aqueous solution is $\leq 11.5$ , or [iii] liquid or semi-solid preparations, the pH of which is $\leq 11.5$<br>Schedule 10 listed |
|--|----------------------------------------------------------------------------------------------------------------------------------------------------------------|

**Australian Industrial Chemicals Introduction Scheme (AICIS)**

| Component                       | Australian Industrial Chemicals Introduction Scheme (AICIS) | Additional information |
|---------------------------------|-------------------------------------------------------------|------------------------|
| Ethyl alcohol - 64-17-5         | Present                                                     | -                      |
| Potassium hydroxide - 1310-58-3 | Present                                                     | -                      |

**Australian - Illicit Drug Precursors/Reagents Substance List**

This product does not contain any substance(s) on the Illicit Drug Precursors/Reagents list.

**Chemicals of Security Concern**

This product does not contain any substance(s) listed on the voluntary National Code of Practice for Chemicals of Security Concern

**National pollutant inventory** Subject to reporting requirements

| Component               | National pollutant inventory      |
|-------------------------|-----------------------------------|
| Ethyl alcohol - 64-17-5 | 10 tonne/yr. Threshold category 1 |

**Prohibition or notification/licensing requirements**

Shown below are details of specific prohibition/notifications or licencing requirements when they apply.

This product does not contain any substance(s) subject to Prohibition, Authorization or Restriction.

**International Inventories**

| Component           | AICS | NZIoC | EINECS    | ELINCS | TSCA | DSL | NDSL | PICCS | ENCS | ISHL | IECSC | KECL     |
|---------------------|------|-------|-----------|--------|------|-----|------|-------|------|------|-------|----------|
| Ethyl alcohol       | X    | X     | 200-578-6 | -      | X    | X   | -    | X     | X    | X    | X     | KE-13217 |
| Potassium hydroxide | X    | X     | 215-181-3 | -      | X    | X   | -    | X     | X    | X    | X     | KE-29139 |

**Legend:** X - Listed. '-' - Not Listed. **KECL** - NIER number or KE number (<http://ncis.nier.go.kr/en/main.do>)

**International Regulations**

**Ozone Depletion Potential** This product does not contain any known or suspected substance

**Persistent Organic Pollutant** This product does not contain any known or suspected substance

**Rotterdam Convention (PIC)** Not applicable

**Basel convention on the control of transboundary movements of hazardous wastes and their disposal**

Take note that wastes may be subject to export, import, or transit controls pursuant to the Basel convention and/or local regulations implementing the Basel convention.

| Component                       | Basel Convention (Hazardous Waste) | Australian Hazardous Waste Act - Categories of Wastes to Be Controlled |
|---------------------------------|------------------------------------|------------------------------------------------------------------------|
| Ethyl alcohol - 64-17-5         | Annex I - Y42                      | Y42 except Halogenated solvents                                        |
| Potassium hydroxide - 1310-58-3 | Annex I - Y35                      | Y35 solid or solution                                                  |

| Component           | CAS No    | OECD HPV | Restriction of Hazardous Substances (RoHS) | Seveso III Directive (2012/18/EC) - Qualifying Quantities for Major Accident Notification | Seveso III Directive (2012/18/EC) - Qualifying Quantities for Safety Report Requirements |
|---------------------|-----------|----------|--------------------------------------------|-------------------------------------------------------------------------------------------|------------------------------------------------------------------------------------------|
| Ethyl alcohol       | 64-17-5   | Listed   | Not applicable                             | Not applicable                                                                            | Not applicable                                                                           |
| Potassium hydroxide | 1310-58-3 | Listed   | Not applicable                             | Not applicable                                                                            | Not applicable                                                                           |

**Authorisation/Restrictions according to EU REACH**

| Component           | REACH (1907/2006) - Annex XIV - Substances Subject to Authorization | REACH (1907/2006) - Annex XVII - Restrictions on Certain Dangerous Substances | REACH Regulation (EC 1907/2006) article 59 - Candidate List of Substances of Very High Concern (SVHC) |
|---------------------|---------------------------------------------------------------------|-------------------------------------------------------------------------------|-------------------------------------------------------------------------------------------------------|
| Potassium hydroxide | -                                                                   | Use restricted. See item 75. (see link for restriction details)               | -                                                                                                     |

<https://echa.europa.eu/substances-restricted-under-reach>

## Section 16 - Other Information

**Legend**

|                                                                                                      |                                                                                                                              |
|------------------------------------------------------------------------------------------------------|------------------------------------------------------------------------------------------------------------------------------|
| <b>AICS</b> - Australian Inventory of Chemical Substances                                            | <b>NZIoC</b> - New Zealand Inventory of Chemicals                                                                            |
| <b>TSCA</b> - United States Toxic Substances Control Act Section 8(b) Inventory                      | <b>EINECS/ELINCS</b> - European Inventory of Existing Commercial Chemical Substances/EU List of Notified Chemical Substances |
| <b>DSL/NDL</b> - Canadian Domestic Substances List/Non-Domestic Substances List                      | <b>ENCS</b> - Japanese Existing and New Chemical Substances                                                                  |
| <b>IECSC</b> - Chinese Inventory of Existing Chemical Substances                                     | <b>KECL</b> - Korean Existing and Evaluated Chemical Substances                                                              |
| <b>PICCS</b> - Philippines Inventory of Chemicals and Chemical Substances                            | <b>CAS</b> - Chemical Abstracts Service                                                                                      |
| <b>TWA</b> - Time Weighted Average                                                                   | <b>ACGIH</b> - American Conference of Governmental Industrial Hygienists                                                     |
| <b>IARC</b> - International Agency for Research on Cancer                                            | Predicted No Effect Concentration (PNEC)                                                                                     |
| <b>ICAO/IATA</b> - International Civil Aviation Organization/International Air Transport Association | <b>IMO/IMDG</b> - International Maritime Organization/International Maritime Dangerous Goods Code                            |
| <b>MARPOL</b> - International Convention for the Prevention of Pollution from Ships                  | <b>ADG</b> Australian Code for the Transport of Dangerous Goods by Road and Rail                                             |
| <b>NZS 5433:2012</b> - Transport of Dangerous Goods on Land                                          | <b>OECD</b> - Organisation for Economic Co-operation and Development                                                         |
| <b>LD50</b> - Lethal Dose 50%                                                                        | <b>LC50</b> - Lethal Concentration 50%                                                                                       |
| <b>EC50</b> - Effective Concentration 50%                                                            | <b>ATE</b> - Acute Toxicity Estimate                                                                                         |
| <b>WEL</b> - Workplace Exposure Limit                                                                | <b>RPE</b> - Respiratory Protective Equipment                                                                                |
| <b>DNEL</b> - Derived No Effect Level                                                                | <b>NOEC</b> - No Observed Effect Concentration                                                                               |
| <b>POW</b> - Partition coefficient Octanol:Water                                                     | <b>BCF</b> - Bioconcentration factor                                                                                         |
| <b>vPvB</b> - very Persistent, very Bioaccumulative                                                  | <b>PBT</b> - Persistent, Bioaccumulative, Toxic                                                                              |
| <b>VOC</b> - (Volatile Organic Compound)                                                             |                                                                                                                              |

**Key literature references and sources for data**

<https://echa.europa.eu/information-on-chemicals>  
Suppliers safety data sheet, Chemadviser - LOLI, Merck index, RTECS

**Classification and procedure used to derive the classification for mixtures according to Regulation (EC) 1272/2008 [CLP]:**

|                              |                       |
|------------------------------|-----------------------|
| <b>Physical hazards</b>      | On basis of test data |
| <b>Health Hazards</b>        | Calculation method    |
| <b>Environmental hazards</b> | Calculation method    |

**Training Advice**

Chemical hazard awareness training, incorporating labelling, Safety Data Sheets (SDS), Personal Protective Equipment (PPE) and hygiene.

Use of personal protective equipment, covering appropriate selection, compatibility, breakthrough thresholds, care, maintenance, fit and standards.

First aid for chemical exposure, including the use of eye wash and safety showers.

Chemical incident response training.

Fire prevention and fighting, identifying hazards and risks, static electricity, explosive atmospheres posed by vapours and dusts.

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|------------------|-----------------|
| Revision Date    | 18-Nov-2022     |
| Revision Summary | Not applicable. |

**This Safety Data Sheet (SDS) is prepared in accordance to and complies with the requirements of Safe Work Australia - Work Health and Safety Regulations (WHS Regulations).**

**Disclaimer**

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**End of Safety Data Sheet**